

Problem Statement – First Deliverable

Basic Information

- **Team Number:** Group Three (3)
- **Project Title:** Assessing the Interplay Between Forestry Dynamics and Climate Change in Syria (1990–2025)
- **Climate Challenge:** Forestry and Terrestrial Ecosystems / Climate Change Impact & Resilience

Problem Statement

Syria has experienced notable shifts in climatic conditions alongside severe environmental pressures and land-use transitions over recent decades. Changes in forest cover and tree-covered areas alter carbon sink capacities and localized climate resilience, while rising temperatures (mean, max, and min) and fluctuating precipitation patterns accelerate forest degradation. Understanding the bidirectional relationship between forestry dynamics (such as forest land cover, tree cover loss, wood production, and forest emissions) and climate parameters (temperature and precipitation timeseries) is vital for regional environmental sustainability. This project investigates how historical trends in forest land cover, tree canopy area, and forestry emissions correlate with long-term climate variations in Syria from 1990 to 2025. By integrating CRU climate timeseries with FAOSTAT forestry and land-use datasets, we aim to quantify these interactions and evaluate how forest cover dynamics reflect and influence regional climate change trends.

Main Research Question

To what extent do historical changes in Syria’s forest cover and forestry emissions (1990–2025) correlate with long-term trends in regional temperature and precipitation?

Datasets Used

1. **CRU TS v4.0x High-Resolution Climate Timeseries** (cru-x0.5_timeseries.xlsx)
 - *Source:* Climatic Research Unit (CRU), University of East Anglia
 - link:
https://cckpapi.worldbank.org/api/v1/cru-x0.5_timeseries_tasmax,tas,pr_timeseries_annual_1901-2025_mean_historical_cru_ts4.10_mean/SYR?_format=json
2. **FAOSTAT Forestry, Land Cover, Land Use & Forest Emissions Datasets** (FAOSTAT_data_*.csv)
 - *Source:* Food and Agriculture Organization of the United Nations (FAOSTAT)
 - links: (land use) [FAOSTAT](#) , (land cover) [FAOSTAT](#) ,(emissions from forests) [FAOSTAT](#) ,(forsetry production) [FAOSTAT](#) .

Expected Output

Python statistical analysis and multi-panel visualisations (time-series trend lines, 10-year rolling averages, correlation matrices, threshold detection, and period comparisons) demonstrating the quantitative relationship between forest cover dynamics, wood production, forest emissions, and climate parameters in Syria.